

SECTION B

Option A – Alternating Currents

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	Choosing correct equation (ωBAN) (1) Correct answer = 421 [V] (1)	1	1		2	1	
		(ii)	Choosing correct equation ($BAN \cos \omega t$) (1) Answer = 0 or close to 0 (1) Award 1 mark for answer of 1.34	1	1		2	1	
		(iii)	421 [V] ecf from (i)		1		1	1	
		(iv)	Period is 20 ms so 5 ms quarter cycle (1) ωt is 90° or equivalent (1) Expect max emf and zero flux (1) Alternative: ωt is 90° or equivalent (1) $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$ (1) Expect max emf and zero flux (1)			3	3		
	(b)	(i)	Choosing correct equation i.e. $\frac{1}{2\pi} \sqrt{\frac{1}{LC}}$ (1) Answer = 103 k[Hz] (1)	1	1		2	1	